

Labsheet - 2

```
import cv2
import numpy as np
import matplotlib.pyplot as plt

# Load image
image = cv2.imread("images/input.jpg")

if image is None:
    raise FileNotFoundError("Could not find images/input.jpg")

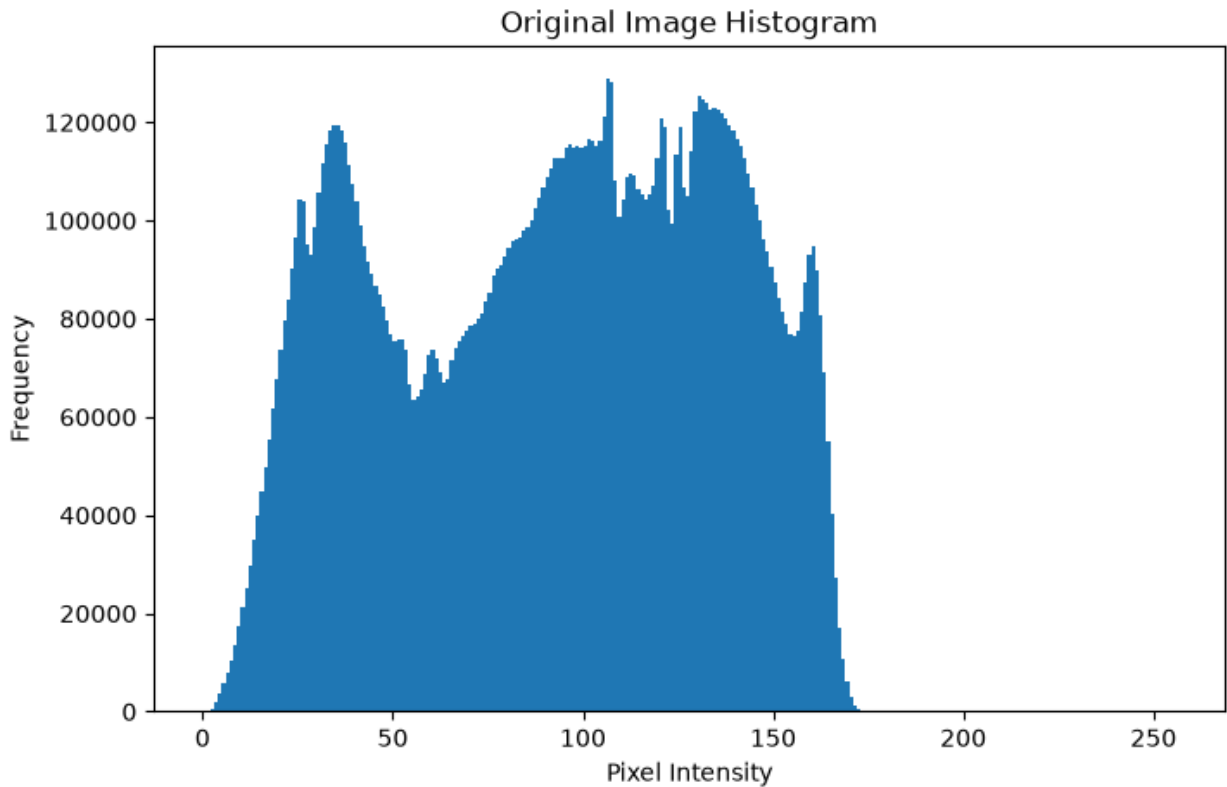
# Convert to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(8, 5))
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

Original Grayscale Image



```
plt.figure(figsize=(8, 5))
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Image Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")
plt.show()
```



```
min_val = np.min(gray)
max_val = np.max(gray)

stretched = ((gray - min_val) / (max_val - min_val) *
255).astype(np.uint8)

plt.figure(figsize=(8, 5))
plt.imshow(stretched, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis("off")
plt.show()
```

Contrast Stretched Image



```
equalized = cv2.equalizeHist(gray)

plt.figure(figsize=(8, 5))
plt.imshow(equalized, cmap="gray")
plt.title("Histogram Equalized Image")
plt.axis("off")
plt.show()
```

Histogram Equalized Image



```
clahe = cv2.createCLAHE(  
    clipLimit=2.0,  
    tileGridSize=(8, 8)  
)  
  
clahe_image = clahe.apply(gray)  
  
plt.figure(figsize=(8, 5))  
plt.imshow(clahe_image, cmap="gray")  
plt.title("CLAHE Enhanced Image")  
plt.axis("off")  
plt.show()
```

CLAHE Enhanced Image



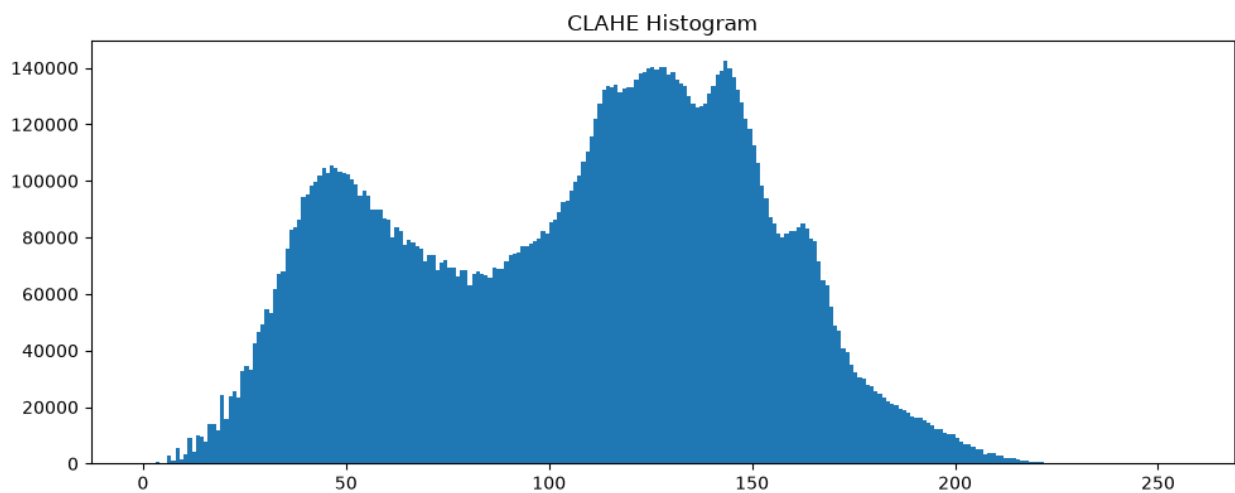
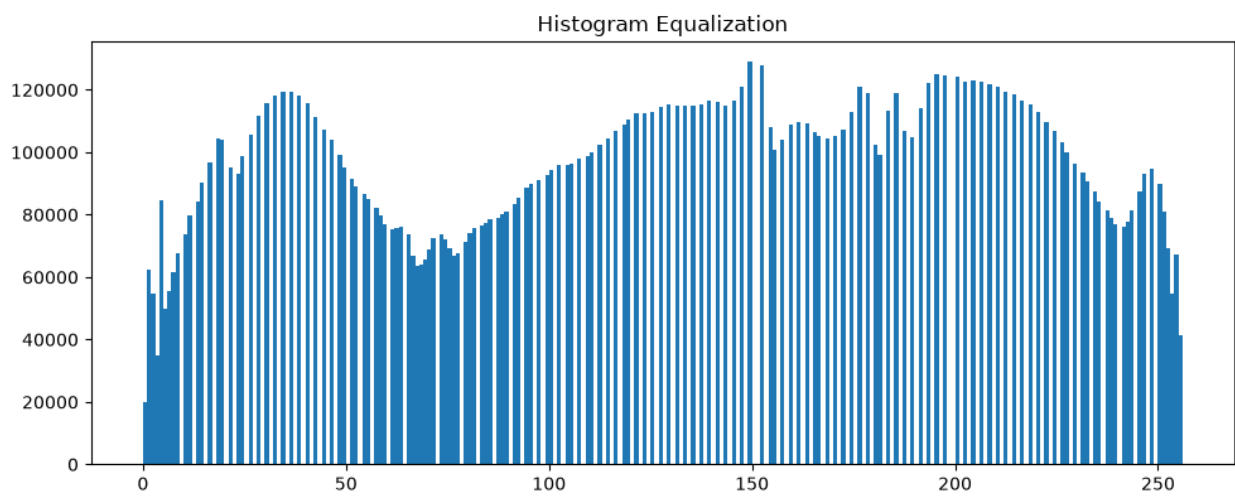
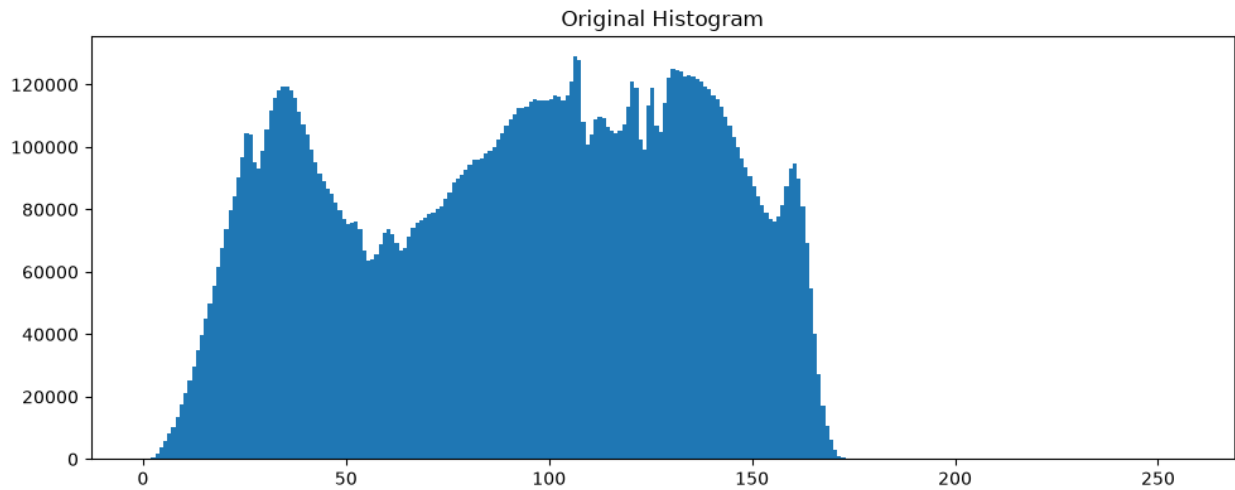
```
fig, axes = plt.subplots(3, 1, figsize=(10, 12))

axes[0].hist(gray.ravel(), bins=256, range=[0, 256])
axes[0].set_title("Original Histogram")

axes[1].hist(equalized.ravel(), bins=256, range=[0, 256])
axes[1].set_title("Histogram Equalization")

axes[2].hist(clahe_image.ravel(), bins=256, range=[0, 256])
axes[2].set_title("CLAHE Histogram")

plt.tight_layout()
plt.show()
```



```
# Load the second image
image2 = cv2.imread("images/input2.jpg")

if image2 is None:
    raise FileNotFoundError("Could not find images/input2.jpg")
```

```
# Convert to grayscale
gray2 = cv2.cvtColor(image2, cv2.COLOR_BGR2GRAY)

# Display original second image
plt.figure(figsize=(8, 5))
plt.imshow(gray2, cmap="gray")
plt.title("Second Image - Grayscale")
plt.axis("off")
plt.show()
```

Second Image - Grayscale



```
equalized2 = cv2.equalizeHist(gray2)

plt.figure(figsize=(8, 5))
plt.imshow(equalized2, cmap="gray")
plt.title("Second Image - Histogram Equalization")
plt.axis("off")
plt.show()
```

Second Image - Histogram Equalization



```
clahe2 = cv2.createCLAHE(  
    clipLimit=2.0,  
    tileGridSize=(8, 8)  
)  
  
clahe_image2 = clahe2.apply(gray2)  
  
plt.figure(figsize=(8, 5))  
plt.imshow(clahe_image2, cmap="gray")  
plt.title("Second Image - CLAHE")  
plt.axis("off")  
plt.show()
```

Second Image - CLAHE



```
fig, axes = plt.subplots(1, 3, figsize=(15, 5))

axes[0].imshow(gray2, cmap="gray")
axes[0].set_title("Second Image - Original")
axes[0].axis("off")

axes[1].imshow(equalized2, cmap="gray")
axes[1].set_title("Second Image - Histogram Equalization")
axes[1].axis("off")

axes[2].imshow(clahe_image2, cmap="gray")
axes[2].set_title("Second Image - CLAHE")
axes[2].axis("off")

plt.tight_layout()
plt.show()
```

Second Image - Original



Second Image - Histogram Equalization



Second Image - CLAHE



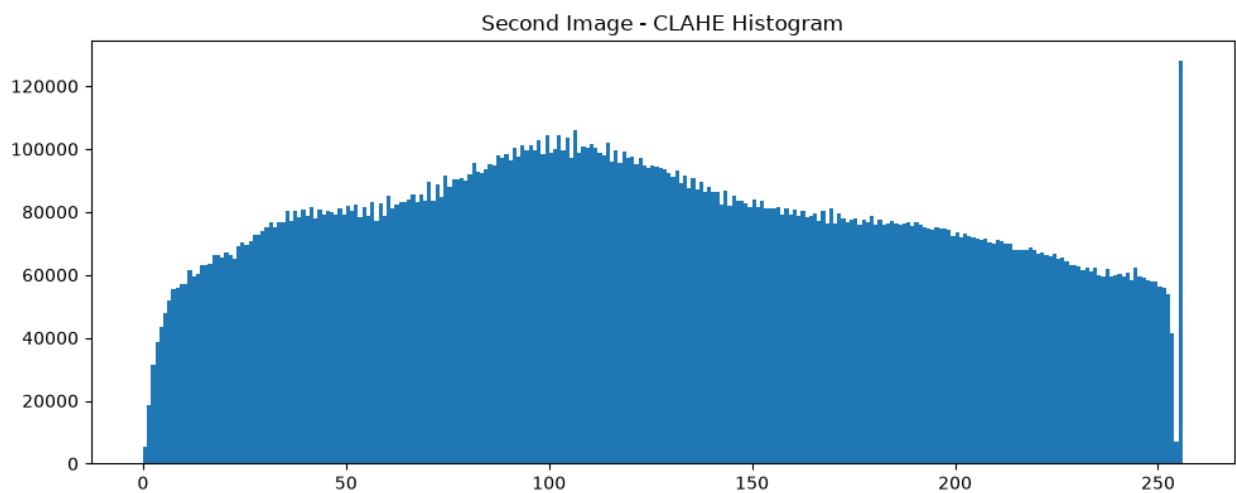
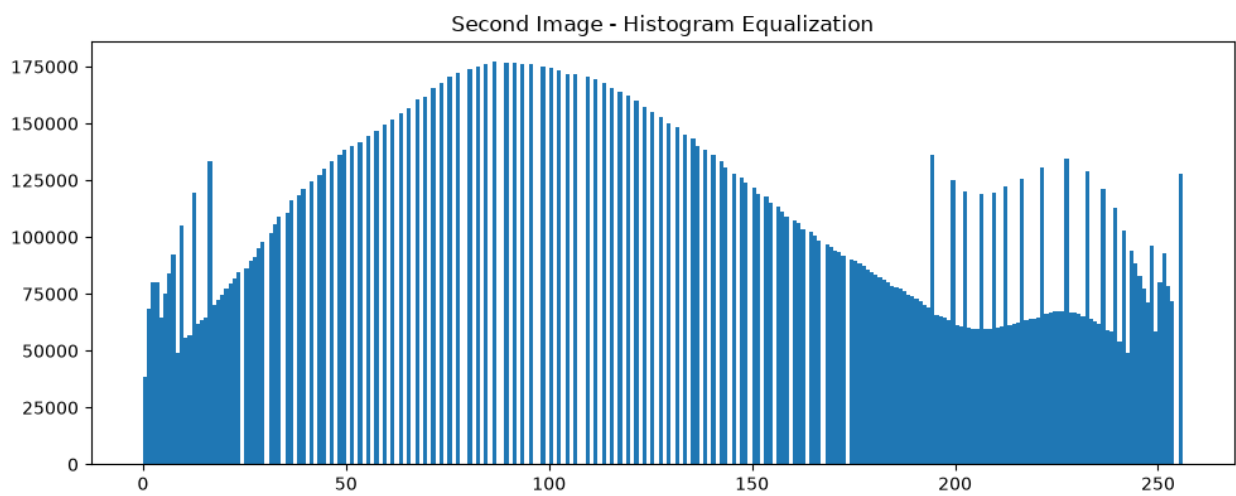
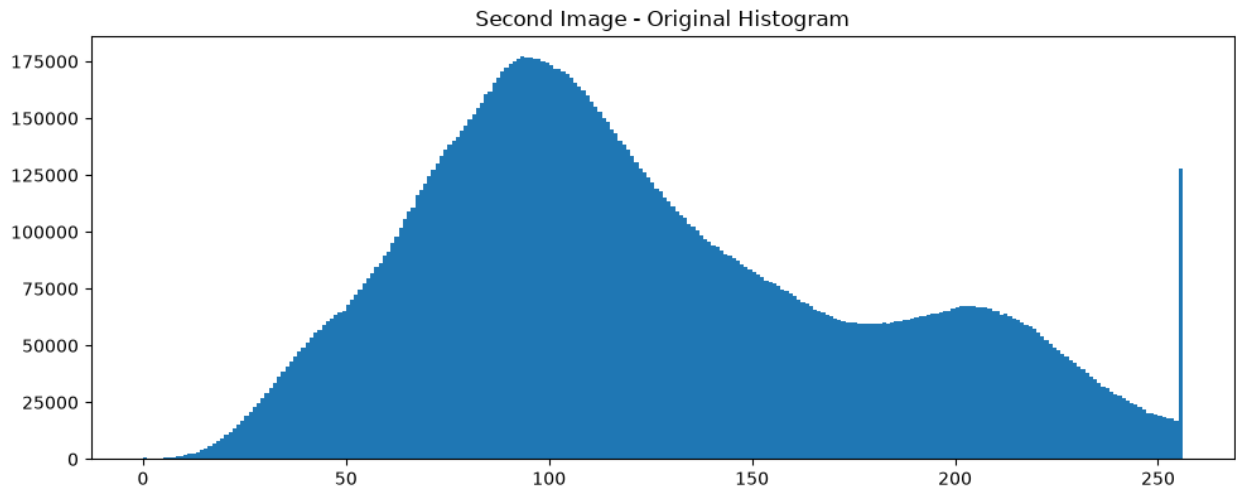
```
fig, axes = plt.subplots(3, 1, figsize=(10, 12))

axes[0].hist(gray2.ravel(), bins=256, range=[0, 256])
axes[0].set_title("Second Image - Original Histogram")

axes[1].hist(equalized2.ravel(), bins=256, range=[0, 256])
axes[1].set_title("Second Image - Histogram Equalization")

axes[2].hist(clahe_image2.ravel(), bins=256, range=[0, 256])
axes[2].set_title("Second Image - CLAHE Histogram")

plt.tight_layout()
plt.show()
```



```
fig, axes = plt.subplots(2, 3, figsize=(15, 10))
```

```
# First image
```

```
axes[0, 0].imshow(gray, cmap="gray")
```

```
axes[0, 0].set_title("Image 1 - Original")
```

```

axes[0, 0].axis("off")

axes[0, 1].imshow(equalized, cmap="gray")
axes[0, 1].set_title("Image 1 - Histogram Equalization")
axes[0, 1].axis("off")

axes[0, 2].imshow(clahe_image, cmap="gray")
axes[0, 2].set_title("Image 1 - CLAHE")
axes[0, 2].axis("off")

# Second image
axes[1, 0].imshow(gray2, cmap="gray")
axes[1, 0].set_title("Image 2 - Original")
axes[1, 0].axis("off")

axes[1, 1].imshow(equalized2, cmap="gray")
axes[1, 1].set_title("Image 2 - Histogram Equalization")
axes[1, 1].axis("off")

axes[1, 2].imshow(clahe_image2, cmap="gray")
axes[1, 2].set_title("Image 2 - CLAHE")
axes[1, 2].axis("off")

plt.tight_layout()
plt.show()

```

Image 1 - Original

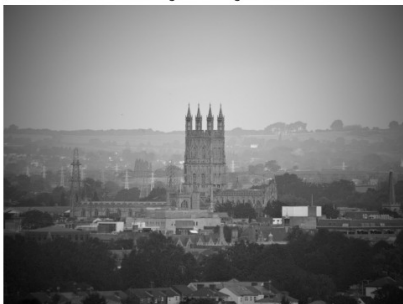


Image 1 - Histogram Equalization

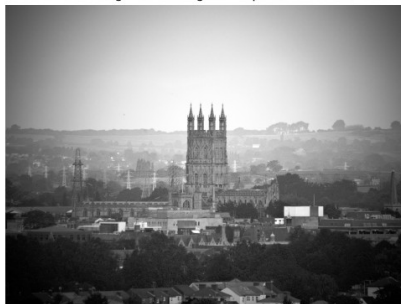


Image 1 - CLAHE



Image 2 - Original



Image 2 - Histogram Equalization



Image 2 - CLAHE



Observations

1. Contrast stretching expands the range of pixel intensities and improves the overall contrast of the image.
2. Histogram Equalization redistributes pixel intensities over a wider range, making details more visible in images with poor contrast.
3. CLAHE improves local contrast by applying histogram equalization to small regions of the image. It generally provides better local detail enhancement while limiting excessive contrast amplification.
4. The visual improvement depends on the image. Histogram Equalization can enhance overall contrast effectively, while CLAHE is useful when illumination varies across different regions.
5. From the histogram comparison, the enhanced images show a broader and more distributed intensity range than the original images.
6. Contrast enhancement can improve the visibility of important features and can therefore support later computer vision tasks such as feature extraction, segmentation, object detection, and recognition.

Conclusion

In this experiment, contrast stretching, Histogram Equalization, and CLAHE were implemented using Python and OpenCV. The original and enhanced images and their histograms were compared. The results show that histogram-based enhancement techniques can improve image visibility and contrast. CLAHE provides localized contrast enhancement and can be useful for images with varying illumination. These techniques are therefore important preprocessing methods in computer vision applications.

Labsheet - 2

1. What is image contrast, and why is it important in image processing?

Answer: Image contrast is the difference in intensity between the bright and dark regions of an image. A high-contrast image has a wide range of intensity values, while a low-contrast image has intensity values concentrated within a narrow range.

Contrast is important because it determines how clearly objects, edges, and details can be distinguished. Improving contrast makes image features more visible and can help subsequent computer vision tasks such as feature extraction, segmentation, object detection, and recognition.

2. Explain the concept of an image histogram. What information does it provide?

Answer: An image histogram is a graphical representation of the distribution of pixel intensity values in an image. For a grayscale image, the horizontal axis represents intensity values, usually from **0 to 255**, while the vertical axis represents the number of pixels having each intensity.

A histogram provides information about the **brightness and contrast** of an image. It can show whether an image is predominantly dark, bright, or has a narrow or wide distribution of intensity values. Histogram analysis is useful for determining whether contrast enhancement is required.

3. Differentiate between Histogram Stretching and Histogram Equalization.

Answer:

Histogram Stretching	Histogram Equalization
Expands the existing intensity range of an image.	Redistributes pixel intensities to produce a more uniform histogram.
Usually maps the minimum and maximum intensities to the full available range.	Uses the cumulative distribution of pixel intensities for redistribution.
Preserves the general distribution of intensities more directly.	Can significantly change the intensity distribution.
Useful when an image uses only a small portion of the available intensity range.	Useful when pixel intensities are unevenly distributed and stronger contrast enhancement is needed.

In this experiment, **contrast stretching** was used to improve the dynamic range, while **Histogram Equalization** was used to redistribute intensity values and enhance contrast.

4. What is Contrast Limited Adaptive Histogram Equalization (CLAHE)? How does it differ from standard Histogram Equalization?

Answer: CLAHE stands for **Contrast Limited Adaptive Histogram Equalization**. It is a local contrast enhancement technique that divides an image into small regions, called tiles, and performs histogram equalization separately in those regions.

The main difference is that standard Histogram Equalization operates on the **whole image globally**, whereas CLAHE works on **local regions**. CLAHE also limits excessive contrast

amplification using a clip limit. As a result, CLAHE can improve local details while reducing the risk of over-enhancing noise.

This makes CLAHE particularly useful for images with **varying illumination**, as required in this experiment.

5. Why is histogram equalization commonly applied before feature extraction and image segmentation?

Answer: Histogram equalization is commonly applied before feature extraction and segmentation because it can improve image contrast and make important features more visible.

When the intensity distribution is poor or concentrated in a narrow range, edges and object boundaries may be difficult to distinguish. Histogram equalization can spread the intensity values over a wider range, making these structures easier to detect.

Therefore, improved contrast can provide better input for subsequent computer vision operations such as **feature extraction and image segmentation**.

6. Mention three real-world applications where histogram equalization is widely used.

Answer: Three real-world applications are:

1. **Medical imaging** — improving the visibility of structures and details in medical images.
2. **Satellite and remote-sensing images** — enhancing features in satellite imagery.
3. **Low-light photography and surveillance** — improving visibility in poorly illuminated images.

These application areas are consistent with the experiment's emphasis on medical images, satellite images, and low-light photographs.

7. What are the limitations of global histogram equalization?

Answer: Global Histogram Equalization has several limitations:

- It treats the **entire image as one region**, so it cannot adapt to local variations in illumination.
- It may produce **excessive contrast** in some areas.
- It can make **noise more visible**, particularly in nearly uniform regions.

- It may produce unnatural-looking results when different parts of an image have very different brightness levels.

Because of these limitations, local methods such as **CLAHE** can be more suitable for images with varying illumination.

8. How does contrast enhancement improve the performance of object detection and recognition systems?

Answer: Contrast enhancement improves the visibility of important image features such as **edges, boundaries, textures, and objects**. When these features become clearer, computer vision algorithms can detect and distinguish objects more effectively.

Better contrast can therefore improve the quality of information supplied to later stages of a vision system, including **feature extraction, object detection, segmentation, and recognition**. The lab specifically identifies improved feature visibility as an important outcome of histogram-based enhancement.

9. Compare histogram-based enhancement techniques with brightness adjustment methods.

Answer:

Histogram-Based Enhancement	Brightness Adjustment
Changes the distribution of pixel intensities to improve contrast.	Primarily shifts pixel intensities to make an image brighter or darker.
Can improve both overall and local contrast depending on the technique.	Does not necessarily improve contrast.
Includes techniques such as Histogram Equalization and CLAHE.	Usually involves adding or subtracting a fixed intensity value.
Can reveal details hidden due to poor contrast.	May make an image brighter while leaving low contrast unchanged.

Therefore, histogram-based techniques are generally more effective when the main problem is **poor contrast**, whereas brightness adjustment is more appropriate when the main problem is simply that the image is too dark or too bright.

10. Why is CLAHE preferred for medical imaging and low-light image enhancement?

Answer: CLAHE is preferred because it performs **local contrast enhancement** instead of applying the same transformation to the entire image. This allows it to improve details in different regions even when illumination varies across the image.

CLAHE also uses a **contrast limit** to reduce excessive contrast amplification, which can help prevent unwanted enhancement of noise. Therefore, it is particularly useful for medical and low-light images where preserving local details and handling uneven illumination are important.

The lab specifically asks students to observe CLAHE's performance on images with **varying illumination** and highlights medical and low-light images as relevant application areas.

Final Conclusion

The experiment demonstrated that **contrast stretching, Histogram Equalization, and CLAHE** can improve the visual quality and information content of low-contrast images. Histogram Equalization provides global contrast enhancement, while CLAHE provides localized enhancement that is better suited to images with varying illumination. These techniques are useful preprocessing methods for subsequent computer vision tasks.